

2023

(Session : 2022-25)

(Paper ID : 11203)

Time : 3 hours

Full Marks : 80

Candidates are required to give their answers in their own words as far as practicable.

The questions are of equal value.

Answer any five questions.

1. What is Array? Explain different types of array with diagram. Also write a program to use them.
2. What do you mean by recursion? Explain different types of recursion. Differentiate looping and recursion.
3. Write a program to multiply two matrices.
4. What is linked list? Explain different types of linked list with neat diagrams. How linked list is advantageous over the array?
5. What is binary tree? Define and prove the different properties of binary tree.
6. What is Queue? Explain different type of operations performed on a queue.
7. What is doubly linked list? Write functions to create, add, delete, display and count the elements of doubly linked list.
8. Write a program to create a binary search tree. Write the recursive functions for display it in pre-order, in-order and post order.
9. Construct a binary tree whose pre-order and in-order traversal are as follows:
(a) Post-order: G D B H I E F C A
(b) In-order: D G B A H E I C F
10. Differentiate any two of following with help of diagram:
(a) Singly Linked List and Doubly Linked List
(b) Stack and Queue
(c) Linear and non-linear data structure